Data-Driven Rockfall Assessment using Synthetic Training and CNN–XGBoost Integration

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Abstract. Rockfall poses a persistent risk to life and infrastructure, demanding proactive, site-specific forecasting. We propose a Hybrid AI framework that fuses convolutional neural networks for quantitative analysis of rock-surface imagery with an XGBoost model ingesting geospatial covariates like rainfall, pore pressure, and temperature to estimate an early-warning probability of slope stability. To address data scarcity and enable controlled experimentation, we construct a comprehensive synthetic dataset spanning diverse geomaterial textures and environmental regimes. The fused model is evaluated using the F1 score and the area under the ROC curve. On a held-out validation set, the framework attains an accuracy of 0.99 and an AUC of 0.99, indicating strong discriminative performance and low false-alarm propensity. These results highlight the promise of multimodal learning for operational rockfall early warning and contribute a reproducible path toward data-driven geohazard assessment.

Keywords: Rockfall Prediction · Geotechnical Engineering · Slope Stability · Artificial Intelligence (AI) · Machine Learning (ML) · Hybrid AI Model · Deep Learning (DL) · Convolutional Neural Network (CNN) · Xtreme Gradient Boosting (XGBoost) · Crack Detection · Synthetic Dataset · Open-Pit Mines · Early Warning Systems.

1 Introduction

Rockfalls are among the most critical and dangerous geological hazards. They are a constant threat to humans and wildlife, causing infrastructural damage, thus impacting the economy. The people who work in dynamic environments such as open-pit mines are at great risk to their lives, where the rock stability is unpredictable. While the traditional methods exist, which are good at analysing rock stability, they struggle to understand complex and non-linear patterns in the geospatial features, especially in dynamic triggers, thus limiting their accuracy in proactive risk management tasks [3]. A critical interval is present in

analysing the rock deformation factors, and most of the approaches focus on the broader geospatial and environmental factors such as Slope Angle, Aspect, Curvature, Lithology, Soil type, etc., rather than exploring the need to analyse the fractures in rocks [28]. To fill this gap, this research contributes to developing a hybrid Artificial Intelligence (AI) framework that fuses modern Deep Learning (DL) methods for image analysis of rock surfaces with a gradient boosting Machine Learning (ML) technique to accurately predict early rockfall cases. Our proposed solution integrates the ability of Convolutional Neural Networks (CNN) that generate a CrackScore from the images of rock surfaces with an Extreme Gradient Boosting (XGBoost) model for geospatial feature modelling, which gives a final single prediction for rockfalls. This results in a multi-modal architecture that combines the geospatial data with the crack analysis from CNN into an XGBoost model to predict a comprehensive rockfall susceptibility score. This hybrid framework is trained on a rule-based novel synthetic dataset, which was able to demonstrate exceptional performance. Our research not only focuses on developing an AI framework for geohazard assessment but also aims to provide a solution to revolutionise the proactive risk management and early warning systems, especially in sensitive areas like open-pit mines.

2 Literature Review

Over a period of time, the engineers have relied on traditional methods and techniques such as the Limit Equilibrium Method (LEM) or field-based rating systems like the Rockfall Hazard Rating System (RHRS) [4, 10, 14]. An extensive work has been completed since 2009. A significant amount of work was published in the time period of 2021 to 2024, followed by two state-of-the-art review papers in 2025. This review is primarily divided into three sub-sections, with each section discussing the work done in that particular approach. The approaches include systems for real-time and post-event detection, large-scale susceptibility mapping systems and at lastly, hybrid and integrated systems for early warning. After that, a fourth section is presented to explore some of the datasets from the literature review, which greatly influenced our feature selection and dataset generation.

2.1 Real-Time and Post-Event detection from sensor data

This section focuses on the papers that detect rockfalls as they happen or after the event. Various modern AI techniques have focused on the important task of rockfall detection from high-resolution sensor data. Some of the methods involve using the Light Detection and Ranging (LiDAR) to create 3D point cloud change datasets of rock slopes and then identifying the changes in them between successive scans. DL models like PointNet are used to automatically classify the 3D point cloud change clusters as rockfalls or non-rockfalls and treat it as a 3D computer vision classification problem [8]. One of the most crucial problems in this area is the data imbalance, where the true rockfalls are very few compared

to other changes. Another research aimed to solve this problem by building an intelligent framework that tested 16 resampling techniques across 11 different ML models like DT, SVM, RF, XGBoost, etc. [35] and found out that the XGBoost model is extremely efficient in such an imbalanced classification task. A technique was demonstrated for monitoring the long transportation corridors without even disrupting traffic using the mobile terrestrial LiDAR, proving that it is accurate enough for geotechnical analysis [20]. A real-time rockfall tracking system was built using the YOLOv5 DL model to detect and track the falling rocks, and with this model, they achieved a mean Average Precision (mAP) of 0.845 [22].

2.2 Large-Scale susceptibility mapping with machine learning

This section specifically focuses on the research that aimed at predicting landslides before they happen over a large geographical area. Along with the rockfall detection, there is another study that focuses on predicting rockfall susceptibility maps. A total of six ML models were trained and compared on the data from the Western Ghats of India, combining a detailed inventory of 184 mapped landslide events with geo-environmental factors such as slope, elevation, rainfall, soil type, etc. The Random Forest (RF) model was found to be the best at creating susceptibility maps with an AUC of 0.88, and the topographic factors such as slope, TRI and elevation were reported as the highest influential features for susceptibility [28]. A similar experiment was also carried out in the Yunnan Province, China, and additionally used the Frequency Ratio (FR) method to create a Landslide Susceptibility Map (LSM). Five ML classifiers were trained, out of which the RF showed the best performance, which gave an accuracy of 0.906 [17]. A Multilayer Perceptron (MLP) was trained with a rich dataset consisting of 15 triggering factors, including ten common factors such as precipitation, slope angle, lithology, etc., along with five geo-engineered features, namely Rock Quality Designation (RQD), Slope Quality Index (Qslope), RHRS, etc., to create these high-accuracy susceptibility maps for Alborz Province, Iran and achieved the highest accuracy of 0.81 and AUC of 0.811 [25]. A CNN-BiLSTM-Attention Mechanism (CNN-BiLSTM-AM) model was trained for landslide susceptibility assessment in the diversified terrain of Nyingchi, Tibet. This hybrid model comprises two individual components: a CNN for spatial feature extraction, and a BiLSTM and attention mechanism for capturing long-range dependencies and dynamically assigning weights to relevant inputs. With this hybrid approach, an accuracy of 90.12% was achieved, outperforming six classical ML models, including RF, SVM, AdaBoost, etc. [15].

2.3 The rise of hybrid and integrated systems for early warning

There is a shift towards developing hybrid and integrated systems that fuse multiple data sources and various models, which has shown an improvement in the accuracy and performance of the ML systems. These ML systems are often developed by fusing predictive models with detection models for building

the AI framework for early warning systems. A Hybrid Early Warning System (HEWS) was successfully developed by integrating two different models, which are a logistic regression model for prediction and a real-time computer vision model for detection, which gave an overall performance of 97.9% [2]. Similarly, an IoT-based AI framework was proposed that fuses a DL model for prediction with a real-time seismic and computer vision model for detection, which contributed to increasing the accuracy of 86% in the case of a single model to an impressive 98.8% with this hybrid framework [1]. Another AI framework was proposed that aims towards fusing a CNN for extracting terrain signatures, vegetation dynamics, soil deformation patterns and climate changes from the high-resolution satellite images for a classification task with other ML algorithms such as SVM, XGBoost, RF for processing the geospatial features like elevation, soil type, rainfall, etc. The model showed exceptional performance and was deployed on a Flask-based web interface for real-time risk assessment [33]. A high-quality dataset of 246 potential landslides in Jingchuan County, China, was built by integrating SBAS-InSAR deformation tracking with object-oriented Landsat image classification. A total of three different models were trained, namely XGBoost, CNN and Blending-XGBoost-CNN. The Blending-XGBoost-CNN outperformed the other two standalone models with the highest AUC and precision of 0.912 [23].

2.4 Review of datasets in rockfall prediction literature

The following section summarise the characteristics of the dataset that are used in influential studies. In one research article, a study on rainfall induced landslides was conducted in Yunnan Province, China. A total of five different ML models were trained, which include XGBoost, KNN, SVM, LR, and RF and found out that RF is the best performing model with accuracy and AUC above 0.9 [17]. Another study where a site-specific dataset was built for Alborz Province in Iran. A Multilayer Perceptron (MLP) was used as their primary model for training, which gave an AUC of 0.811 [25]. One more study was focused on Rockfall detection, to detect whether the marks on the surface of the land are caused by a rockfall or by any other factor. In this research, they built a unique and detailed dataset from a 5-year-long LiDAR-based monitoring program. With this process, a massive dataset of 8,900 individual 3D change clusters was created. These 8,900 clusters were manually labelled as "rockfall" and "non-rockfall". Two models on this labelled dataset were trained, which are PointNet and PointNet++. These models look at the whole 3D point cloud at once to learn its overall characteristics [29, 30]. After training, they found that the PointNet model performed better with an accuracy of 89% on the Mile 109 site and an impressive 93% on White Canyon West [8]. A detailed summary of these datasets is given in Table 1. The characteristics of the datasets summarised above helped us in selecting the correct and meaningful geospatial input features, along with the novel rules through which we generated the synthetic dataset for the training of the hybrid model.

Table 1: Summary of key datasets.

Data Category	**Primary Goal**	**Sample Size**	**Features**
Historical records from Yunnan Province, China. [17]	Rainfall-induced landslide forecasting and susceptibility mapping [17]	11,610 landslides (FR), 256 dated (train/test), 20 dated (validation), matched non-landslide samples [17]	Elevation, slope, aspect, lithology, land cover, NDVI, soil type, average annual precipitation, event daily rain, antecedent effective rainfall [17]
Historical inventory and field surveys in Alborz, Iran. [25]	Regional rockfall susceptibility with ANN, compared to SVM/DT/RF [25]	57 rockfall events; 157 labelled units (57 susceptible, 100 non-susceptible), 70/30 split [25]	Precipitation, slope aspect, slope angle, weathering, lithology, distance to rivers, faults unsafe radius, landslide-prone areas, distance to roads, distance to cities, RQD, Qslope, GSI, SMR, RHRS [25]
5-year 3D LiDAR monitoring data from British Columbia, Canada. [8]	Automated rockfall detection/ classification from 3D point clouds with DL [8]	8,966 deformation clusters; 3,466 rockfalls; test on 536 clusters; additional site dataset for generalization [8]	Raw 3D point coordinates, global PointNet embeddings, local PointNet++ multi-scale features, implicit volume/shape descriptors [8]

3 Dataset Summary

The most important requirement for developing an efficient ML model is a robust, precise, and relevant dataset. For a problem like rockfall prediction, finding existing datasets with the right features is a significant challenge. In this section, we will introduce our own proposed synthetic dataset, which has been used to train and test our model for rockfall prediction.

3.1 Proposed synthetic dataset for hybrid modelling

For training our proposed model, the biggest challenge was data, as for the prediction of rockfall, we need to train the model on real-life sensor data along with the images of the rocks. We analysed a lot of aspects, and finally defined 9 important features that contribute to the rockfall. These features include Slop Angle, Aspect, Curvature, Displacement, Strain, Rainfall, Temperature, Vibrations, and Pore Pressure. A detailed overview of each feature is given in Table 2, which contains the units, range, and nature of the features. The Slope Angle, Aspect, and Curvature are derived from Digital Elevation Models (DEM). The Displacement, Strain, Vibrations, and Pore pressure are derived from Sensors while the Rainfall and Temperature are derived from weather data.

Table 2: Features used in XGBoost model training.

Feature	**Unit**	**Typical Range**	**Nature**
Slope Angle	Degrees (°)	35 to 85	Continuous
Aspect	Degrees (°)	0 to 360	Cyclical/Categorical
Curvature	Unitless Index	-1 to +1	Continuous
Displacement	mm/week	0 to 20	Continuous
Strain	Microstrain ($\mu\varepsilon$)	0 to 5000	Continuous
Rainfall	Cumulative mm in last 72 hours	0 to 200	Continuous
Temperature	Degrees Celsius (°C)	-10 to 40	Continuous, with a special condition
Vibrations	Peak Particle Velocity (PPV) in mm/s	0 to 50	Continuous
Pore pressure	Kilopascals (kPa)	0 to 100	Continuous
Crack Score	Probability	0.0 to 1.0	Continuous

The main challenges reside in finding this dataset, which contains all of these features along with the images of the rocks. After deep searching, we failed to find the dataset needed to train a model that would be capable of predicting rockfall. To address this challenge, we created our own novel synthetic dataset, which was highly inspired by recent studies that have successfully generated synthetic datasets using probabilistic simulations to train their predictive models for rockfall risk probability [32]. And hence we built our own novel synthetic dataset, which contains all of the features mentioned above.

We found a dataset of concrete surface sample images from kaggle.com [16]. In this dataset, the image data is divided into two classes: one as negative (without a crack) and the other as positive (with a crack). Each class contains 20,000 images, which makes a total of 40,000 images with 227 x 227 pixels with RGB channels. This dataset is generated from 458 high-res images (4032x3024 pixels) applying a method of patch-based classification for automatic crack detection, which was introduced in a prior foundational research [34]. This image dataset was already preprocessed and ready to use for the training. Although this concrete dataset reflects the fracture patterns, hight-contrast edges and discontinuities in the joints but they do not fully capture the textures and real essence of natural rocks. Thus this mismatch may reduce the generalisation ability of the model when used on natural rock surface images.

The structural CrackScore and other geospatial features mentioned in Table 2 were generated based on geologically meaningful rules rather than randomising the dataset. We described six scenarios, and they include Intense Rainfall, Vibration-Induced scenario due to an earthquake or blast shock, Low Freezing Temperature, Progressive Failure, Combination Failure due to multiple high-risk factors, and an All clear scenario, which will not lead to any rockfall. The dataset is designed in such a way that the CrackScore correlates with the slope mechanics and other geospatial features in each of these scenarios. We generated

a total of 20,000 unique samples to provide a comprehensive training dataset for our model. To prevent bias, the dataset was perfectly balanced, containing 10,000 samples representing "High Risk" scenarios (Target Label: 1) and another 10,000 samples representing "Low Risk" scenarios (Target Label: 0). This dataset was programmatically generated using Python 3.10 [11], leveraging the NumPy library [13] for numerical operations and the Pandas library [24] for data structuring and exportation. The complete process of dataset generation is shown in Figure 1.

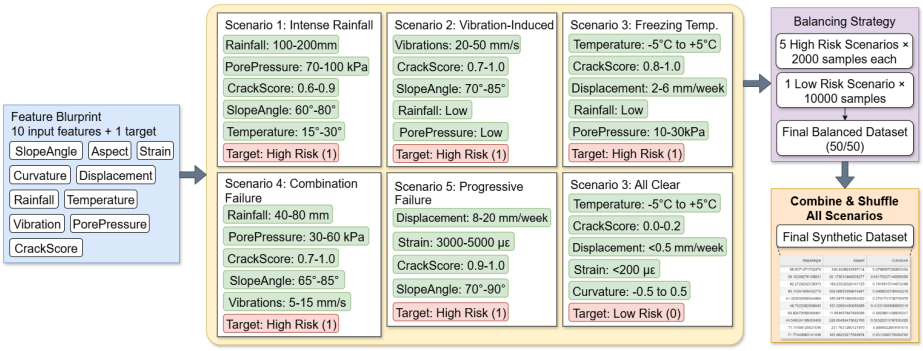


Fig. 1: Process flow of synthetic rockfall dataset generation based on defined scenario rules and feature distributions.

4 Experiment and Results

In this section, we will discuss the training and testing of the two models used, namely CNN and XGBoost, as well as the computational environment employed to train these models. Additionally, we will assess the final performance of the integrated hybrid model pipeline. We aimed to build a hybrid pipeline using these two models to reach a prediction for rockfall. This hybrid approach combining CNN and XGBoost is highly inspired by the modern methodologies that employ DL and ML algorithms and integrate them to amplify the predictive performance [33]. In their research, they also utilised a CNN for processing the satellite images and a Gradient Boosting model for the geospatial data, such as elevation, soil type, and rain intensity.

4.1 Experimental setup

All of our experiments were conducted on a limited computational power workstation, with the specifications and key software libraries detailed in Table 3. The performance of the XGBoost model was evaluated using standard metrics, which

include Accuracy, Precision, Recall, F1-score, and Support derived from the confusion matrix [5, 9, 27]. For evaluating the CNN model, we used two standard metrics, which are Accuracy and AUC [9, 12].

Table 3: Experimental hardware and software specifications.

Component	Specification
Hardware	
GPU	NVIDIA GeForce RTX 3050
Software	
Python Version	3.10.16
Key Libraries	TensorFlow 2.10.1, XGBoost 3.0.5, Scikit-learn 1.6.1, Pandas 2.2.3

4.2 Performance of the hybrid prediction framework

The full flow of the hybrid model from input to output is illustrated in Figure 2. The concrete crack dataset, including 40,000 images, was used to train the CNN model. CNNs are multi-layered deep neural networks that have shown exceptional performance in tasks like image classification, achieving phenomenal performance in complex visual challenges [19]. The dataset was split in an 80-20 ratio, 80% for the training and 20% for the validation. The Adam Optimizer [18] was used for optimizing the weights of the neurons, and the Binary Cross-Entropy loss function was used to penalise the model for adjusting its weights and biases [31]. The model architecture consists of an input layer, which accepts images and rescales the pixel values from [0, 255] to a more manageable [0, 1] range. After the input layer, there are three Convolutional blocks for extracting the important features from the image, with each block having a Conv2D layer with ReLU as the activation function, along with a MaxPooling2D layer. The first block's Conv2D layer has 32 filters of 3x3 size to find the basic patterns. The second block's Conv2D layer has 64 filters of 3x3 size to find even more complex patterns, and the last block's Conv2D layer has 128 filters of 3x3 size to capture even more details. Following these three Convolutional blocks, there comes the Flatten layer, which converts the matrix into a 2D array for the neural network to process for classification. A fully connected dense layer of 128 neurons with ReLU activation function is equipped after the Flatten layer [21], and a dropout of 0.5 is given to prevent overfitting of data. At last, a single neuron is used with a sigmoid activation function to convert the output of the dense layer into 0 or 1 for the CrackScore prediction [21]. This model has a total number of 11,169,089 trainable parameters. The CNN model with this architecture is trained for 15 epochs, which took almost 25 minutes to train on an RTX 3050 GPU.

The model's learning progress over 15 epochs is illustrated in Figure 3. The training and validation accuracy almost converged, indicating successful learning

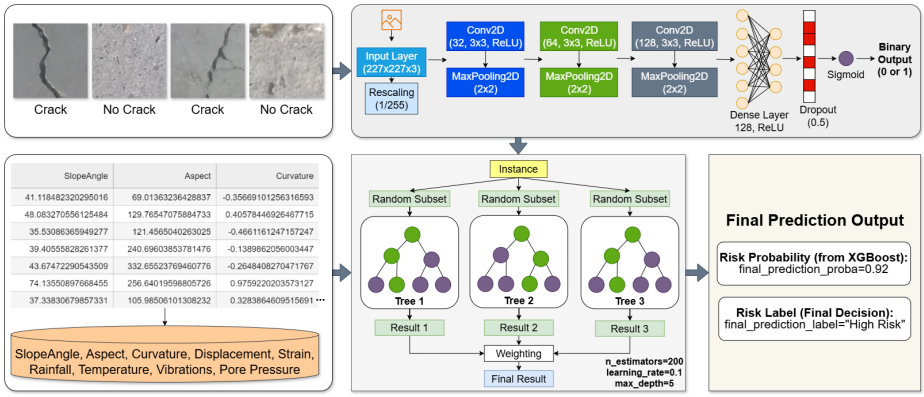


Fig. 2: The proposed hybrid model architecture for rockfall risk prediction.

without significant overfitting. The model achieved a final validation accuracy of 0.9942 and an AUC of 0.9989. The validation loss measured at the end of the training is 0.0221. Secondly, the XGBoost model was used with a total of 200 trees, with each tree having a max depth of 5 nodes. A learning rate of 0.1 was used to make the training process more robust. The XGBoost model stands for Xtreme Gradient Boosting, which uses decision trees at the very core and sequentially combines them [6]. The dataset was split into an 80-20 ratio, 80% for training and 20% for testing. This CrackScore from CNN prediction is appended with the other geospatial data, including SlopeAngle, Aspect, Curvature, Displacement, Strain, Rainfall, Temperature, Vibrations and Pore Pressure, thus forming a comprehensive multi-modal feature matrix. The XGBoost model was trained on 16,000 multi-modal training samples. On the unseen test set of 4,000 samples, the model achieved a classification accuracy of 100%, as depicted in the classification report as shown in Table 4.

Table 4: Classification report for the XGBoost model.

Class	**Prec.**	**Recall**	**F1**	**Support**
Low Risk (0)	1.00	1.00	1.00	2000
High Risk (1)	1.00	1.00	1.00	2000
Accuracy			1.00	4000
Macro Avg	1.00	1.00	1.00	4000
Weighted Avg	1.00	1.00	1.00	4000

The confusion matrix in Figure 4a visually confirms the model's perfect performance, with zero false positives and, critically for a safety application, zero false negatives. An analysis of the model's feature importances was conducted to

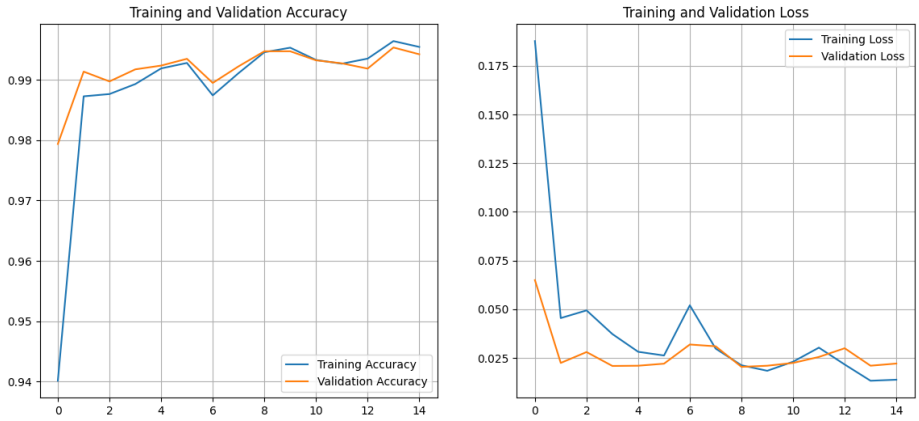
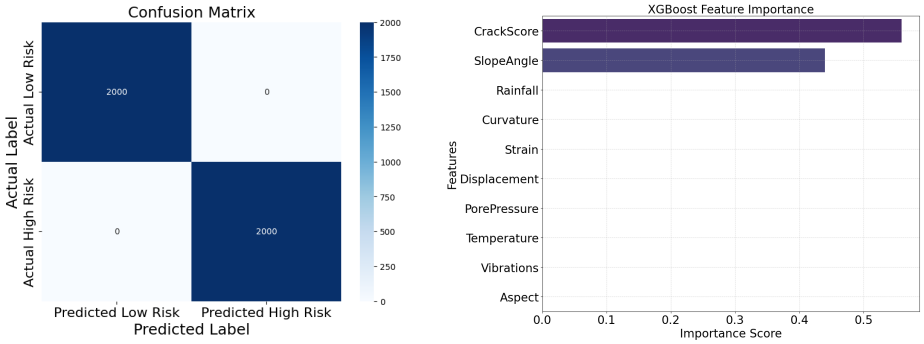


Fig. 3: Training and validation accuracy (left) and loss (right) curves for the CNN crack detection model over 15 epochs.

understand its decision-making process, see Figure 4b. The results indicate that CrackScore and SlopeAngel were the most influential factors, which aligns with established geotechnical principles. It was highlighted that the two primary requirements for the initiation of rockfalls are the discontinuities in the rock joints and the steepness of the slope that directly contribute to the rockfall failure [7]. The RHRS has also stated that the cracks or discontinuities in the rock joints and the geometry of slopes are the highest influential features in the rockfall evaluation framework [26]. Thus, the feature importance scores directly relate to the fundamental rock mechanics.

4.3 Evaluation of the hybrid prediction pipeline

The output of the CNN model, which is a crack score (0 or 1), is used as a feature alongside other geospatial features and fed as an input to the XGBoost model, which then gives an output in the form of final prediction probability and final prediction label. To validate the end-to-end logic of the hybrid system, it was evaluated against six distinct, handcrafted scenarios. The results, shown in Table 5, demonstrate that the model correctly classifies most of the scenarios according to its underlying risk factors. One of the scenarios was labelled for expected prediction as "Low Risk", but the model predicted it as "High Risk", which was the Tricky Case, cracked but stable. The model's prediction here was influenced by two major factors: (i) a very high CrackScore predicted by the CNN model and (ii) an extremely steep SlopeAngle of 60° in the input data. This scenario was named as Tricky Case because although the CrackScore and SlopeAngle are high but there can still be a case where the rock is stable until a little push by any other environmental factor. This was a complex case where the model predicted high potential for rockfall. The robustness and generalisation



(a) (b)

Fig. 4: Model evaluation results for the XGBoost classifier: (a) confusion matrix showing separation between low-risk and high-risk classes, and (b) bar chart of feature importance scores highlighting magnitude as the most influential predictors.

capability of the individual models were separately evaluated using a 5-fold cross-validation. The Accuracy, F1-score and AUC were computed and reported the mean ± deviation for each fold as shown in the Table 6.

Table 5: Hybrid model performance on predefined test scenarios.

Scenario	**Crack Score**	**Final Prediction**	**Expected Prediction**
Intense Rainfall	1.00	High Risk	High Risk
Vibration-Induced	1.00	High Risk	High Risk
Freezing Temperature	1.00	High Risk	High Risk
Progressive Failure	1.00	High Risk	High Risk
Stable Baseline	0.00	Low Risk	Low Risk
Tricky Case (Cracked but Stable)	1.00	High Risk	Low Risk

Table 6: Five-fold cross-validation results for the CNN and XGBoost models.

Model	**Accuracy**	**F1-score**	**AUC**
CNN	0.9906 ± 0.0020	0.9906 ± 0.0021	0.9990 ± 0.0009
XGBoost	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000

5 Conclusion

In this study, we successfully developed a hybrid AI framework that is trained and validated using a novel synthetic dataset, thus providing a considerably good performance and accuracy for predicting the complex rock stability. Our hybrid AI model aimed at developing a pipeline with a CNN at the first step for scoring the rock surface images for cracks, and then this crack score is passed on with other geospatial features like slope angle, rainfall, pore pressure, etc. to an XGBoost ensemble model. The framework was evaluated using metrics like accuracy, F1-score, AUC, etc. The CNN model achieved an outstanding validation accuracy of 0.9942 with an AUC of 0.9989, while the final XGBoost classifier achieved an accuracy of 1.0, precision of 1.0, recall of 1.0, and F1-score of 1.0. This exceptional performance of our hybrid framework marks a milestone in proactive risk management, which can be used in early warning systems to evade the danger to lives in sensitive environments, especially in open-pit mines, where the rockfalls are an unpredictable and constant threat. This hybrid model has a higher inference time because of the CNN feature extraction followed by the XGBoost model prediction. Optimising this pipeline in the framework can help to reduce this inference time for real-time warning systems. In conclusion, this research not only aimed at developing a hybrid AI framework but also established a foundation for developing dynamic early warning systems that can be deployed worldwide.

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